

Rectangular block encoding via state-reflection synthesis

$K = 8$ blocks; phase_bitsize $b = 32$.

Each A_k is an $M \times N$ matrix with possibly non-power-of-2 M, N .

The padded unitary dimension is $D = \text{smallest power of } 2 \geq M + N$.

- If $M \leq N$: 'transposed' synthesis builds the first M rows of a $D \times D$ unitary using M state reflections (each in a D -dim Hilbert space).
- If $M > N$: 'standard' synthesis builds the first N columns of a $D \times D$ unitary using N state reflections (each in a D -dim Hilbert space).

Underlying bloq:

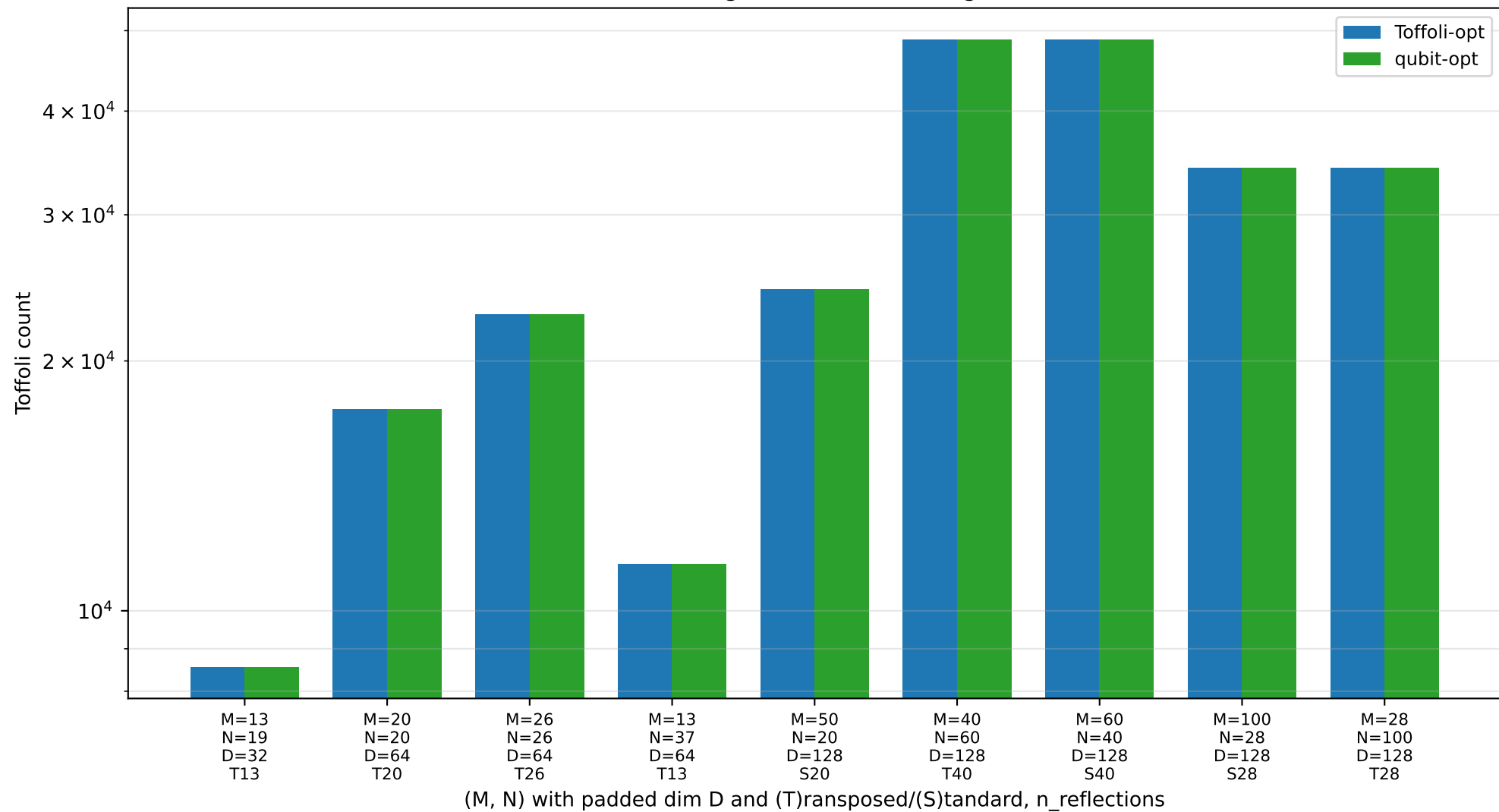
`BlockUnitarySynthesisQROAM(n_blocks=K, n_rows=D, n_reflections=min(M, N)).`

Reflections are loaded by QROAM with the rows (transposed) or columns (standard) of A_k , padded with zeros up to D . No comparator is needed in the circuit; the non-power-of-2 cutoff is enforced purely by the data.

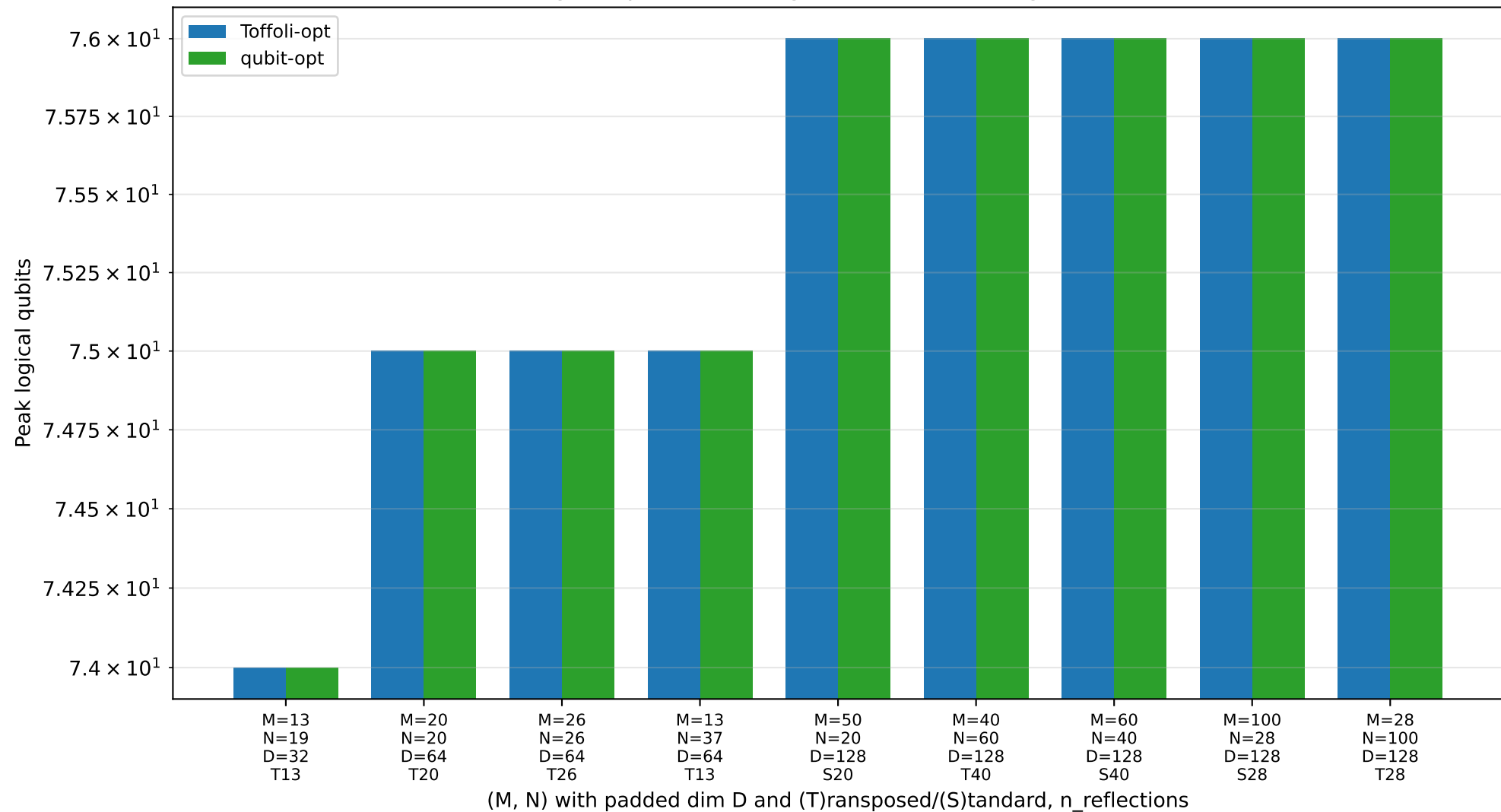
`build_call_graph` and `build_composite_bloq` are both implemented and yield the same Toffoli and (peak) qubit counts via Qualtran's `get_cost_value`.

Each row's `log_block_sizes` (forward + adjoint) is swept over $0..11$ and minimized for (T, Q) lex (Toffoli-opt) and (Q, T) lex (qubit-opt).

Toffoli count: rectangular block encoding ($K=8$, $b=32$)



Peak logical qubits: rectangular block encoding ($K=8$, $b=32$)



Resource counts (K=8, b=32)

M	N	D	n_refl	method	T-opt T	T-opt Q	Q-opt T	Q-opt Q	T-opt lbs	Q-opt lbs
13	19	32	13	transposed	8,554	74	8,554	74	(0, 3)	(0, 3)
20	20	64	20	transposed	17,500	75	17,500	75	(0, 3)	(0, 3)
26	26	64	26	transposed	22,750	75	22,750	75	(0, 3)	(0, 3)
13	37	64	13	transposed	11,375	75	11,375	75	(0, 3)	(0, 3)
50	20	128	20	standard	24,400	76	24,400	76	(0, 3)	(0, 3)
40	60	128	40	transposed	48,800	76	48,800	76	(0, 3)	(0, 3)
60	40	128	40	standard	48,800	76	48,800	76	(0, 3)	(0, 3)
100	28	128	28	standard	34,160	76	34,160	76	(0, 3)	(0, 3)
28	100	128	28	transposed	34,160	76	34,160	76	(0, 3)	(0, 3)